REVISION WORKSHEET

UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

AL KHALIDIYA BOYS SCHOOL

Cycle 3 (10048) | Dibba Al Hisn | Sharjah Eastern

Practice Questions and Worked Answers

Grade 9 Advanced | Module 2 Lesson 1 | Picturing Motion

All 30 questions are followed by answers. Study the reasoning, formula, substitution and units. Equivalent scientifically correct wording is acceptable.

Useful equations and guidance: Equal-time motion diagrams: compare successive positions and their spacing. Elapsed time = number of intervals × interval duration.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 9 book, printed pages 29–31.

P01. An object is moving relative to a reference point when its:

A. Mass changes B. Position changes with time
C. Colour changes D. Temperature is constant

Answer

B. Position changes with time.

P02. A particle model represents an object by:

A. Its detailed surface texture B. Only its mass
C. A point at each recorded position D. A continuous photograph without timing

Answer

C. A point at each recorded position.

P03. In a motion diagram, successive images should usually be separated by:

A. Equal distances regardless of motion B. Random times
C. Different coordinate units D. Equal time intervals

Answer

D. Equal time intervals.

P04. Equal dot spacing along a straight path suggests:

A. Uniform motion B. Speeding up
C. Slowing down D. An instant change of direction

Answer

A. Uniform motion.

P05. Increasing dot gaps at equal time intervals indicate:

A. Decreasing speed B. Increasing speed
C. No motion D. A smaller elapsed time between every image

Answer

B. Increasing speed.

P06. A motion diagram shows positions 0, 4, 8 and 12 m at equal times. What is the spacing between successive positions?

A. 8 m B. 2 m
C. 4 m D. -4 m

Answer

C. 4 m. Successive spacing = 4 − 0 = 4 m; the equal spacing indicates uniform motion.

P07. A camera records 5 successive positions separated by 0.2 s. Calculate the time from the first image to the last.

A. 1.6 s B. 0.4 s
C. -0.8 s D. 0.8 s

Answer

D. 0.8 s. There are 5 − 1 = 4 intervals. Total time = 4 × 0.2 = 0.8 s.

P08. An object travels 6 m forward and then 8 m farther forward along the same line. Find the total distance travelled.

A. 14 m B. 28 m
C. 7 m D. -14 m

Answer

A. 14 m. Distance = 6 + 8 = 14 m.

P09. In a motion diagram, one gap is 4 cm and a later gap is 10 cm. Calculate how much the gap increased.

A. 12 cm B. 6 cm
C. 3 cm D. -6 cm

Answer

B. 6 cm. Increase = later gap − earlier gap = 10 − 4 = 6 cm. With equal time intervals, increasing gaps indicate speeding up.

P10. Repeated readings are 22, 24 and 26 cm. Calculate the range of the readings.

A. 8 cm B. 2 cm
C. 4 cm D. -4 cm

Answer

C. 4 cm. Range = largest − smallest = 26 − 22 = 4 cm.

P11. What is a motion diagram?

Answer

A sequence of images or points showing an object's position at successive times.

P12. Why is a reference object needed when describing motion?

Answer

Motion is a change in position relative to a chosen reference; different references can lead to different descriptions.

P13. When is a particle model appropriate?

Answer

When the object's size, shape and internal motion are not important to the motion being studied.

P14. How does a motion diagram show an object at rest?

Answer

Successive positions coincide, assuming the same reference frame and equal time intervals.

P15. Why must the timing of images be known?

Answer

Without timing, dot spacing alone cannot determine whether changes in spacing reflect speed changes or unequal recording intervals.

P16. A motion diagram shows positions 0, 6, 12 and 18 m at equal times. What is the spacing between successive positions?

Answer

Successive spacing = 6 − 0 = 6 m; the equal spacing indicates uniform motion.

P17. A motion diagram shows positions 0, 8, 16 and 24 m at equal times. What is the spacing between successive positions?

Answer

Successive spacing = 8 − 0 = 8 m; the equal spacing indicates uniform motion.

P18. A camera records 6 successive positions separated by 0.2 s. Calculate the time from the first image to the last.

Answer

There are 6 − 1 = 5 intervals. Total time = 5 × 0.2 = 1 s.

P19. A camera records 7 successive positions separated by 0.2 s. Calculate the time from the first image to the last.

Answer

There are 7 − 1 = 6 intervals. Total time = 6 × 0.2 = 1.2 s.

P20. An object travels 9 m forward and then 12 m farther forward along the same line. Find the total distance travelled.

Answer

Distance = 9 + 12 = 21 m.

P21. An object travels 12 m forward and then 16 m farther forward along the same line. Find the total distance travelled.

Answer

Distance = 12 + 16 = 28 m.

P22. In a motion diagram, one gap is 6 cm and a later gap is 15 cm. Calculate how much the gap increased.

Answer

Increase = later gap − earlier gap = 15 − 6 = 9 cm. With equal time intervals, increasing gaps indicate speeding up.

P23. In a motion diagram, one gap is 8 cm and a later gap is 20 cm. Calculate how much the gap increased.

Answer

Increase = later gap − earlier gap = 20 − 8 = 12 cm. With equal time intervals, increasing gaps indicate speeding up.

P24. Repeated readings are 23, 25 and 27 cm. Calculate the range of the readings.

Answer

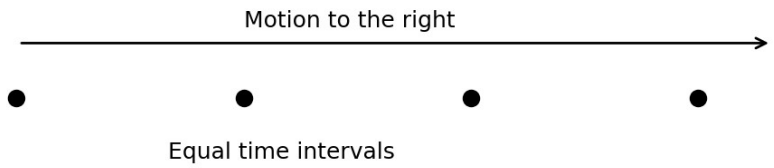
Range = largest − smallest = 27 − 23 = 4 cm.

P25. Repeated readings are 24, 26 and 28 cm. Calculate the range of the readings.

Answer

Range = largest − smallest = 28 − 24 = 4 cm.

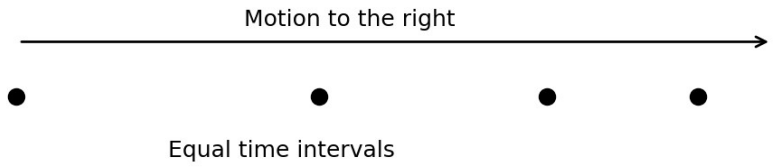
P26. Sketch four successive positions for an object moving right at constant speed. State the time-interval assumption.



Answer

Draw four equally spaced dots from left to right and indicate motion to the right. The dots must represent equal time intervals.

P27. Sketch a motion diagram for an object slowing down while moving right.



Answer

Place successive dots from left to right with progressively smaller gaps, using equal time intervals.

P28. A passenger says a bag on a train seat is stationary; a platform observer says it is moving. Explain.

Answer

Both can be correct. The bag is at rest relative to the train but changes position relative to the platform.

P29. Why can a blurred photograph alone be insufficient for a quantitative motion description?

Answer

It may not identify distinct positions and known time intervals. A calibrated sequence or motion diagram gives more reliable position-time information.

P30. Two diagrams have equal dot gaps but different time intervals between dots. Must their speeds be equal? Explain.

Answer

No. Equal distances can be covered in different times. The diagram with the shorter time interval represents the greater speed.